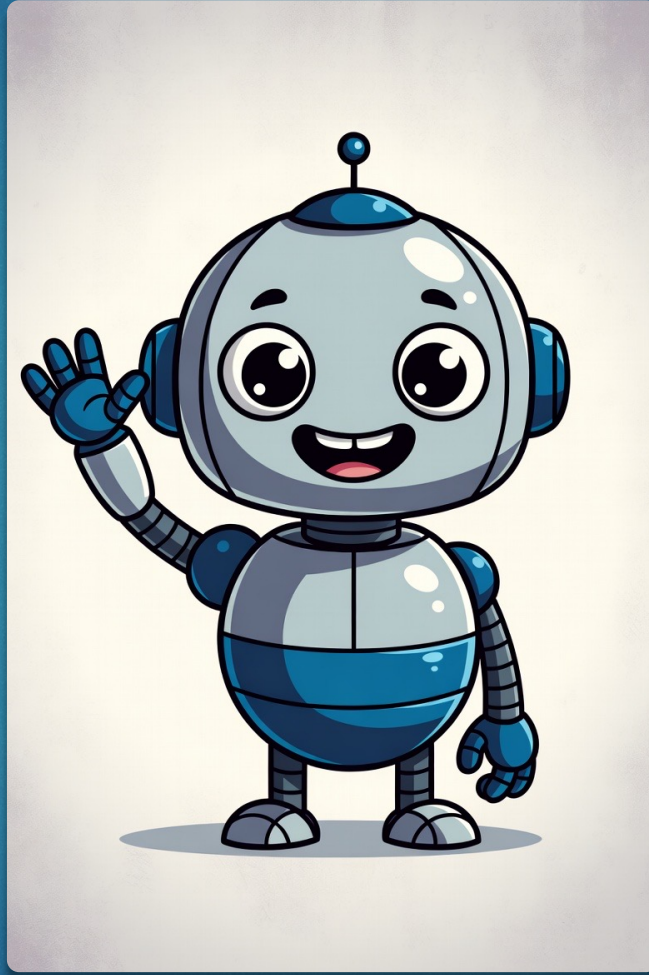




AI Lab

Introduction

QUINTON GILLANDERS
SOFTWARE DEVELOPER THAT LIKES BUILDING COOL APPS

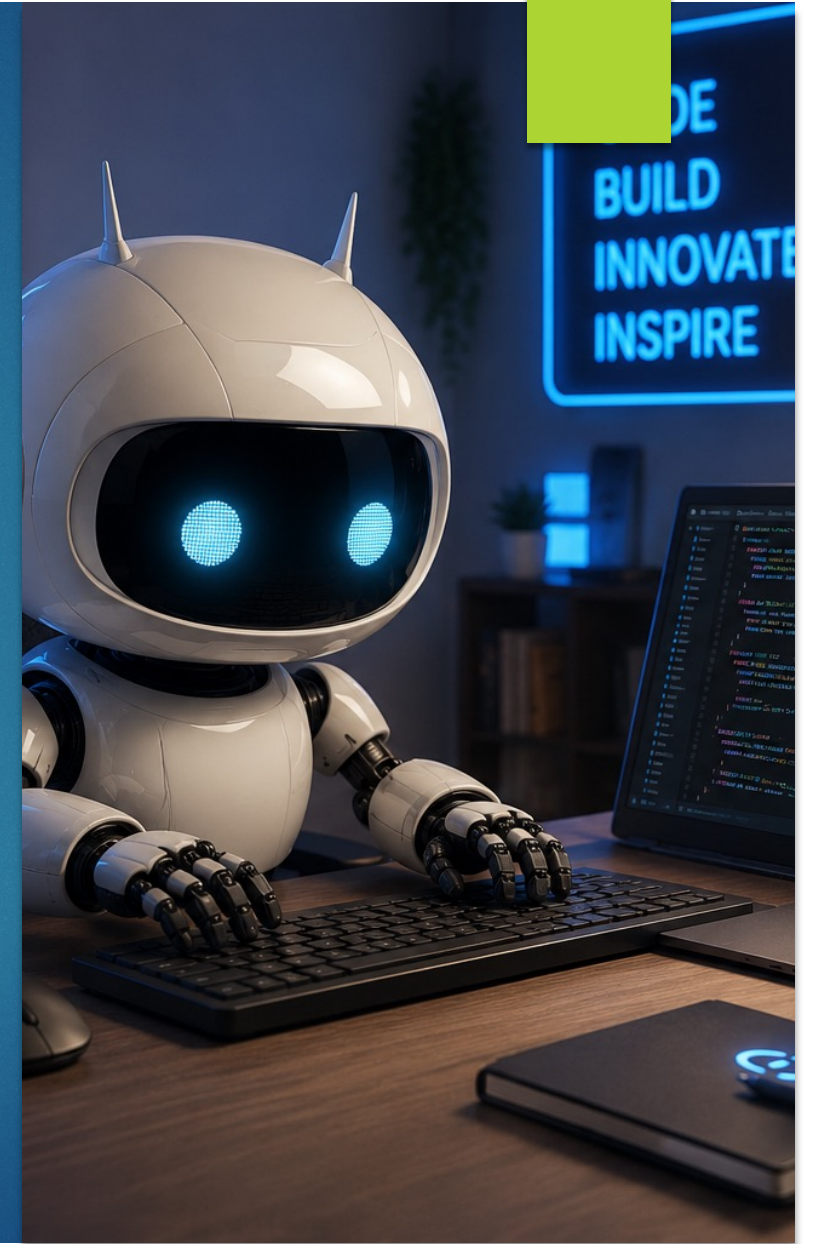


By the end of this lesson, you will:

UNDERSTAND WHAT AI IS (HISTORY, DEFINITION, COMMON USE)

RECOGNIZE AI IN EVERYDAY LIFE (SOCIAL MEDIA, GOOGLE MAPS, NETFLIX, SIRI, CHATGPT, ETC).

BUILD CURIOSITY AND CONFIDENCE IN ENGAGING WITH AI.





Google Maps



Have you
used AI
Before?

Lesson 1 - What is AI?

AI learns from lots of information

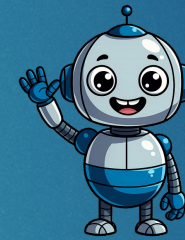
AI can answer questions and solve problems

AI can create stories, pictures and music

While it is helpful, it can also make some mistakes



Human Intelligence VS Artificial Intelligence



Humans	AI
Have Emotions	No Emotions
Creative	Learns from examples
Can use common sense	Follows patterns
Learn from life experiences	Learns from data

How does AI Actually Work?

AI Looks at millions of examples

It finds patterns, and uses those patterns to make predictions

Example:

It will look at thousands of cat photos, learn that they have pointy ears, and then will be able to recognize another cat.



How does AI Actually Work?

When given a task, AI will use the patterns to predict outputs. Some more examples:

- Typing "hav" - autocorrect will predict "have"
- Spotify will recommend songs like the ones you already love
- TikTok and Youtube will show videos based on what you watch and rewatch



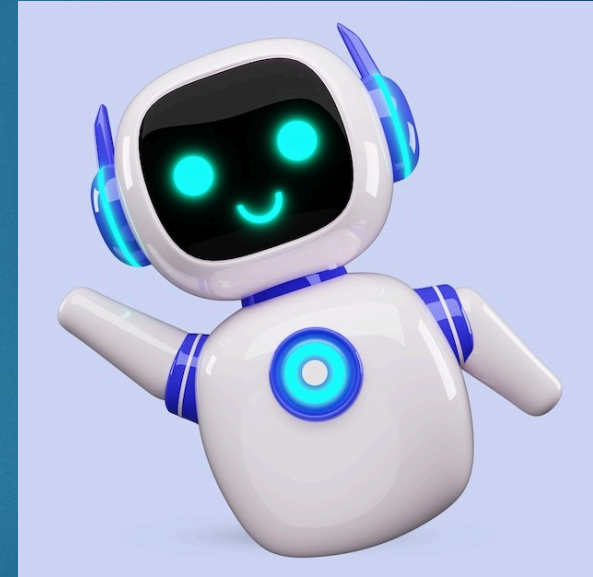
Icebreaker: AI in my pocket

Step 1: List 3 ways you already use AI in your daily life.
(Hint: think about social media, apps, games or devices at home!)

Step 2: Share and compare!
Turn to a partner and share one of your examples.

- Did they come up with the same ones as you?
- Or did they think of something you hadn't noticed?

Also be ready to share one cool example with the class!



Activity 1: Design your own AI!

Invent a brand-new AI tool that could be useful (or fun!) in everyday life. Give it a name, explain what data it would need to be trained on, and describe what it could do.

- 1) What is the name of the AI?
- 2) What data does it learn from?
- 3) How does it work?
- 4) What problem does it solve? And why it's fun!
- 5) Bonus Challenge: Draw a quick sketch of what your AI could look like!



Different Types of AI in daily life!

AI isn't just one thing, it's everywhere. From the moment you wake up and check your phone, to the time you go to bed scrolling through your favorite apps.

Examples:

Netflix recommending a new show

Snapchat applying a funny filter to your face

Your phone suggesting the quickest route to your school

Different Types of AI in daily life!

Common types you see every day:

- Recommendation AI
- Recognition AI
- Language AI
- Predictive AI
- Creative AI

Activity 1 – Debate Corner = Helpful or too smart?

Pick one of the following discussion starts and share your opinions:

1. AI makes life easier
2. AI knows us too well
3. Creative AI is not real creativity
4. AI will eventually replace some jobs – and that's okay.

Each side shares their thoughts for 2 minutes, then swaps roles and argues with the opposite side!

Activity 2 – Spot the AI (Whole class game!)

I will describe different situations, and you must call out whether it uses AI or not!

Whoever gets the most right wins a mini prize or bragging rights!

Activity 1: The good, the bad and the curious (Pairs or groups)

List examples of how AI helps the world and where it could cause problems

The Good	The Bad	The Curious (Qs)
Detect diseases early	Spreads fake information	Can AI have feelings?
Reduces pollution	May replace some jobs	Who controls what AI learns?



Activity 2: If I could use AI to change the world (Individual)

Imagine you could use AI to fix one global problem – hunger, climate change, education, poverty, etc.

Write or draw your idea:

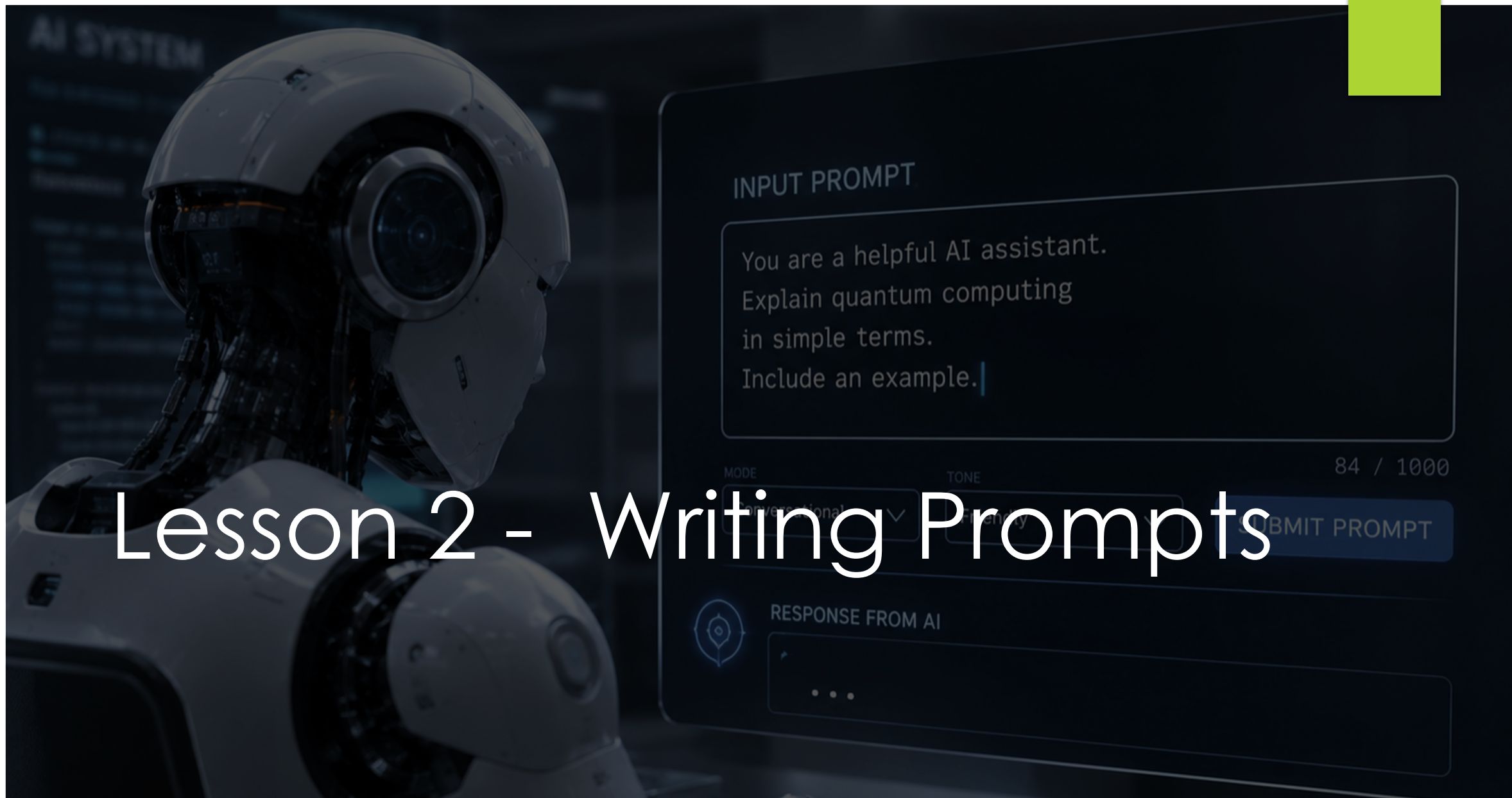
- What problem would you solve?
- What kind of AI would you create to do it?
- Who would benefit the most?

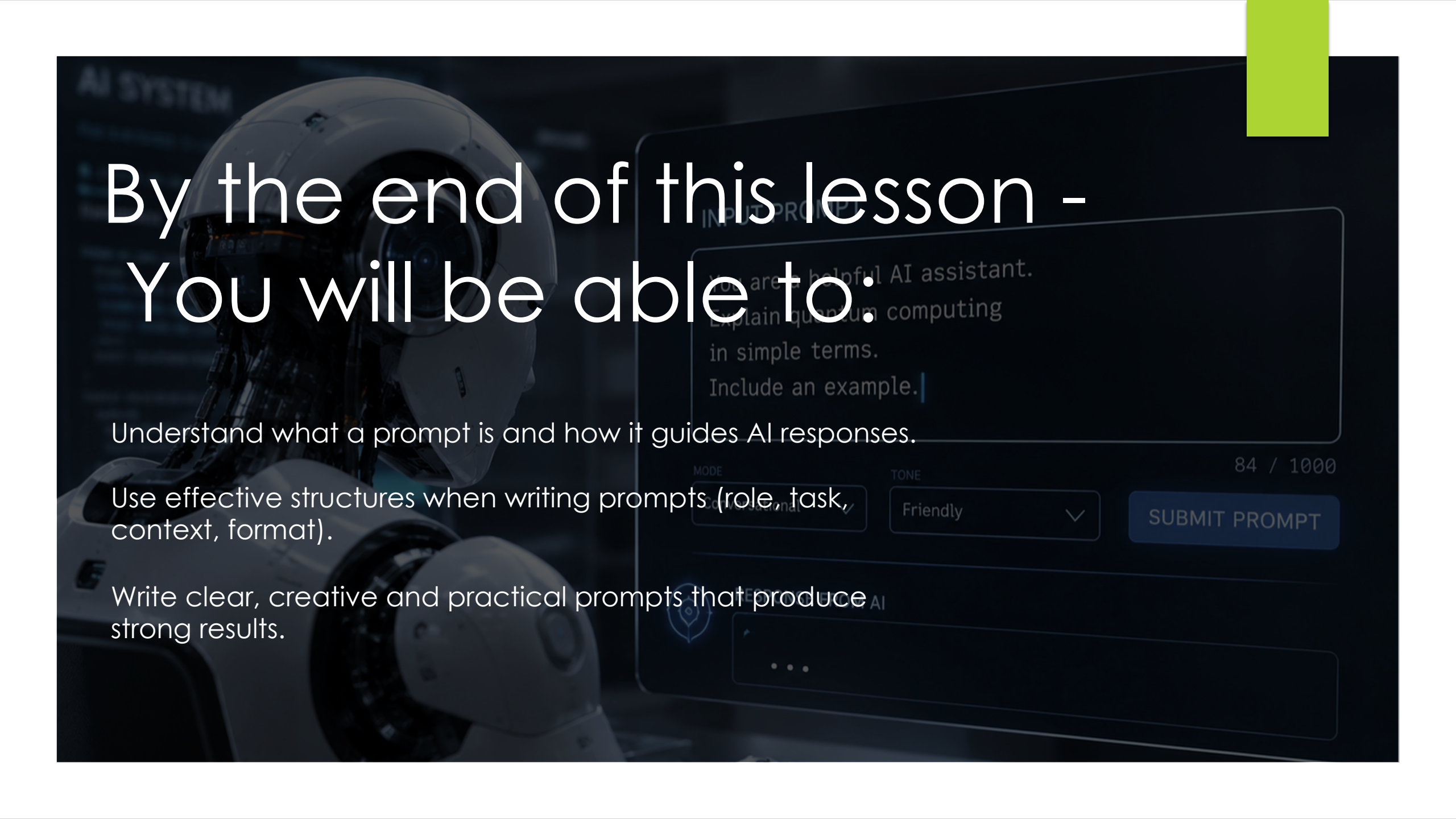
Be creative – there are no wrong answers!



End of
Lesson 1!

Lesson 2 - Writing Prompts



The background of the slide features a dark, futuristic aesthetic. On the left, a white robot head with a camera-like eye is visible. On the right, a digital interface is shown with a text input area containing a prompt. The prompt text is: "You are a helpful AI assistant. Explain quantum computing in simple terms. Include an example." Below the input area, there are dropdown menus for "MODE" (set to "Conversational") and "TONE" (set to "Friendly"). A "SUBMIT PROMPT" button is to the right of these menus. A character count "84 / 1000" is visible. At the bottom of the interface, there is a response area with three dots indicating a loading or processing state. A bright green vertical bar is located in the top right corner of the slide.

By the end of this lesson - You will be able to:

Understand what a prompt is and how it guides AI responses.

Use effective structures when writing prompts (role, task, context, format).

Write clear, creative and practical prompts that produce strong results.

What is a prompt?

A prompt is an instruction or question you give to an AI tool.

It is how we communicate with AI and tell it what we want it to do, how we want it done, and what kind of response we can expect

The AI's output is only as good as your input!



Let's learn what a prompt is! ♥

What is a Prompt?

A prompt is the **instructions** or **question** you give to an AI.
It tells the AI what you want it to do or what information you need.

EXAMPLE:

YOUR PROMPT	AI RESPONSE
Explain photosynthesis in simple terms. 	Photosynthesis is how plants make their own food using sunlight, water, and carbon dioxide. 

TIP:
The **clearer** your prompt, the **better** the AI's response!

What is a prompt?

Imagine if you're asking a friend to draw a picture of you

If you just said, "Draw something", that would mean anything.

Now imagine you ask: "Draw a cute cartoon cat wearing sunglasses and holding an ice cream"



Let's learn what a prompt is!

What is a Prompt?

A prompt is the **instructions** or **question** you give to an AI.
It tells the AI what you want it to do or what information you need.

EXAMPLE:

YOUR PROMPT	AI RESPONSE
Explain photosynthesis in simple terms.	Photosynthesis is how plants make their own food using sunlight, water, and carbon dioxide.

TIP:
The **clearer** your prompt, the **better** the AI's response!

Result



Activity 1 – Prompt Detectives!

The goal for this activity: Spot the most effective prompt.

Below are three prompts AI prompts

"Write about animals"

"Write a fun paragraph about why penguins can't fly"

"Animals info please"

Which is best and why?

How could you improve the other two prompts?



Activity 2 – Draw it with words

Your goal is to visualize how clarity shapes creativity.

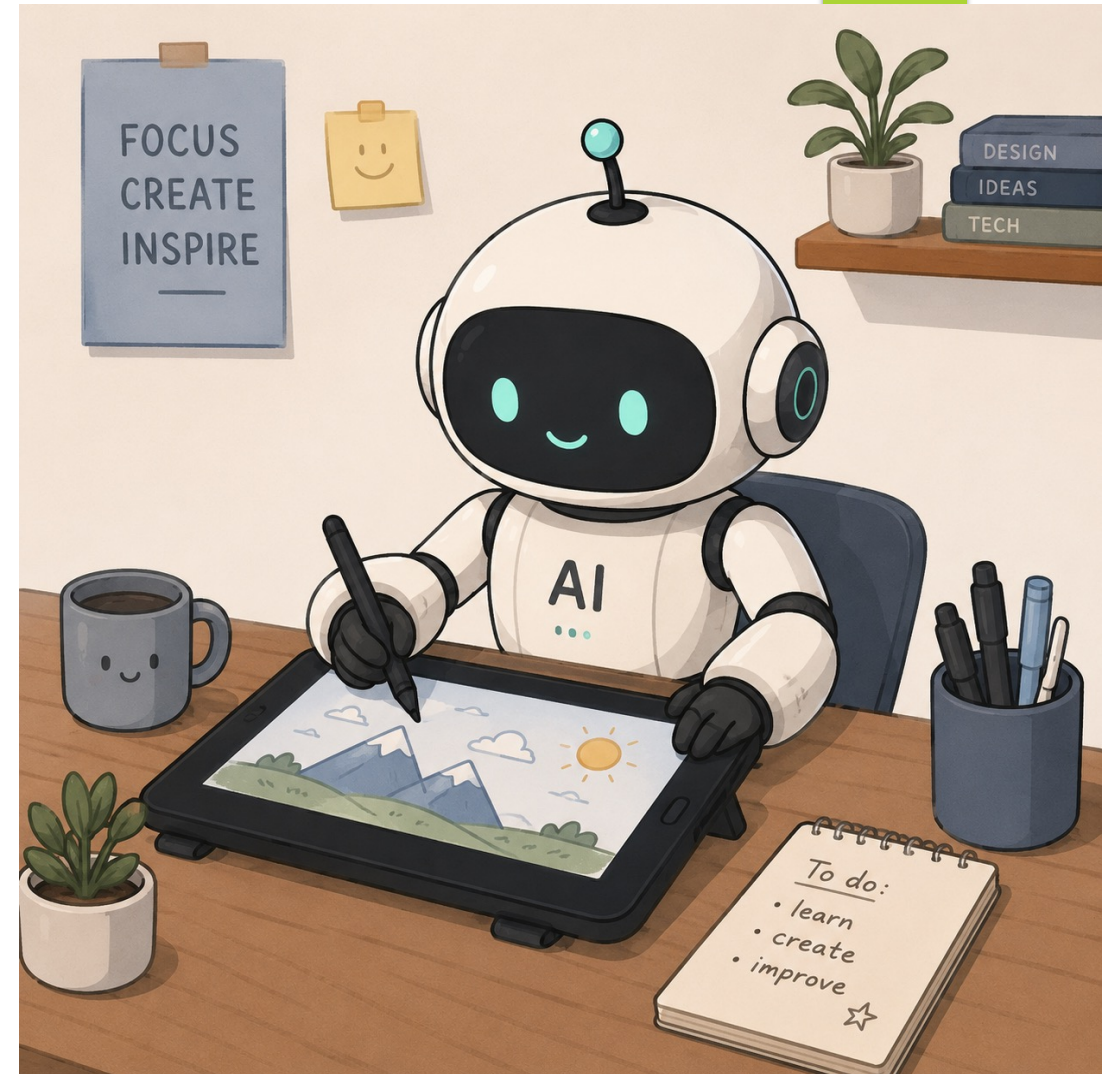
Imagine you're asking AI to "draw" something using words.

Write one short vague prompt and one clear descriptive prompt for the same image.

Examples:

Vague: "Draw something cool"

Specific: "Draw a robot surfing a rainbow wave under a starry sky."



Activity 2 – Draw it with words

Reflection:

What details made the second prompt stronger?



The Prompt Formula

Every strong prompt usually includes four key parts that work together to tell the AI exactly what to do and how to do it

Part	What it means	Example
Role	Tell the AI who it should be or act as	"You are a travel agent..."
Task	Tell it what to do – the main goal or activity	"...write a short itinerary..."
Context	Give background details or information to help the AI tailor its response	"... for a family visiting Japan..."
Format	Explain how you want the answer to look – list, paragraph, table, script, etc.	"... as a bullet-point list."

Why each part matters

When you give AI a role, you're telling it who to become for your request

If you say "You're a teacher" - The AI explains things simply

If you say "You're a scientist" - The AI becomes more technical and detailed.

Some more examples:

"You are a film critic, write a review for Toy Story 5"

"You are a primary school teacher, explain photosynthesis to Year 5 students"

"You are a travel blogger, describe how your trip to Paris was".



Role Roulette

GROUP GAME

Role Roulette

Part 1

Instructions

Your goal is to experiment with how changing the role impacts the AI's tone and response.

I will write the different roles clearly on the whiteboard

- Artist
- Teacher
- CEO
- Wizard
- Astronaut

Example:

"Describe how to make a sandwich"

Artist: "You are an artist describing a sandwich as a masterpiece".

Scientist: "You are a scientist explaining the sandwich process step-by-step".

Role Roulette Part 2 Instructions

Some prompts don't quite hit the mark,
can you fix them?

Here are some weak prompts:

1. "Write about space."
2. "You are a chef. Talk about food"
3. "Make a poem about love"

Your job is to rewrite each one to make it strong by adding all four parts – Role, Task, Context and Format

Example fix:

"You are a science teacher. Explain how recycling helps the planet to a Year 6 class using simple bullet points".



End of
Lesson 2!

Lesson 3 - Advanced Prompting and Refinement

INPUT PROMPT

You are a helpful AI assistant.
Explain quantum computing
in simple terms.
Include an example.

84 / 1000

MODE

Conversational

tone

Friendly

SUBMIT PROMPT



RESPONSE FROM AI

...

By the end of this lesson, you will be able to:

Learn advanced strategies, role assignment,
multi-step prompts, iteration

Compare simple vs advanced prompting

Develop problem solving skills with AI

INPUT PROMPT

You are a helpful AI assistant.
Explain quantum computing
in simple terms.
Include an example.

MODE

Conversational

tone

Friendly

84 / 1000

SUBMIT PROMPT



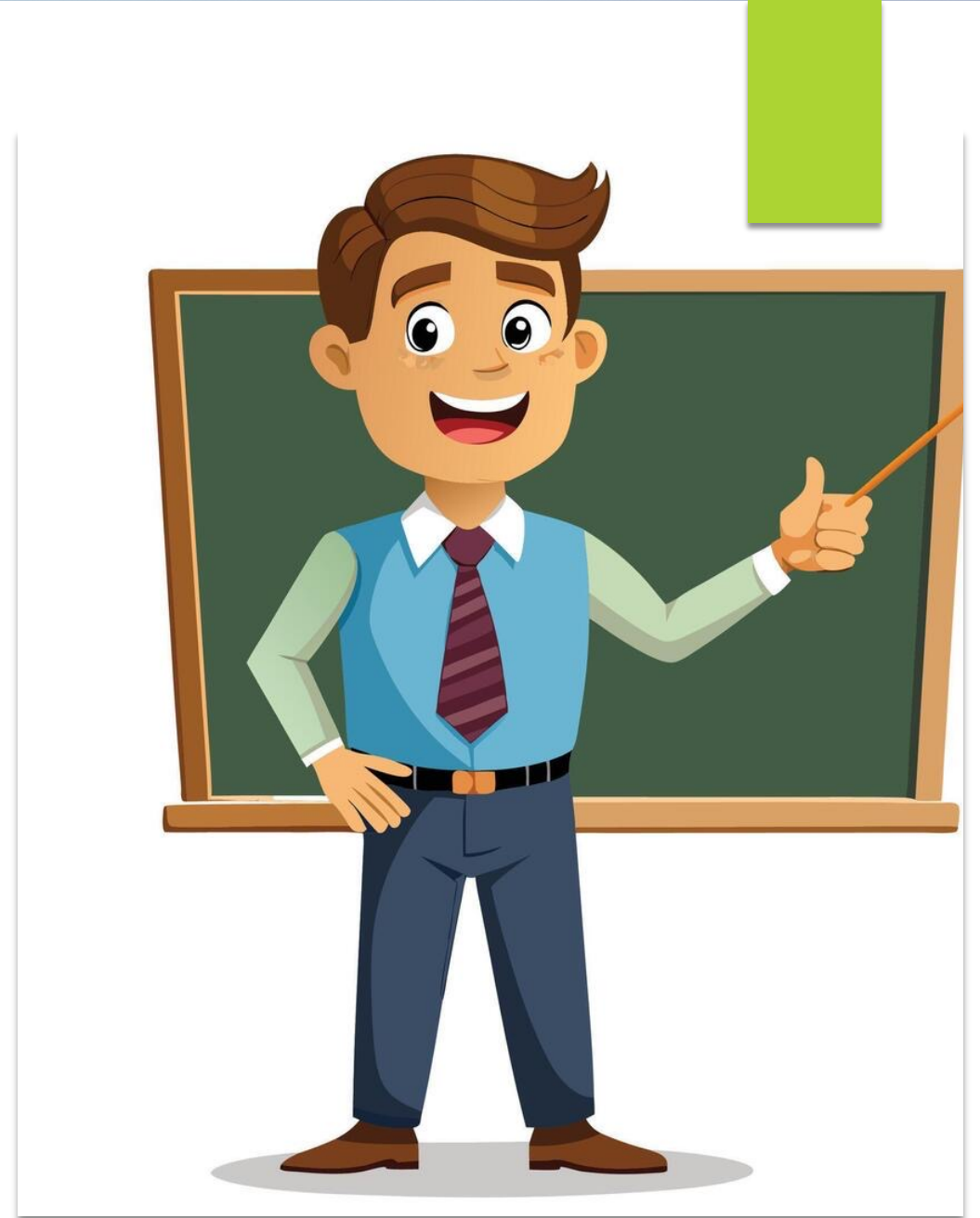
RESPONSE FROM AI

...

Simple Vs Advanced Prompting

Simple Prompt - "Write a story about friendship"

Advanced Prompt: "You are a friendly Year 7 teacher, write a warm story about friendship using 100 words".



Technique 1 – Role Assignment

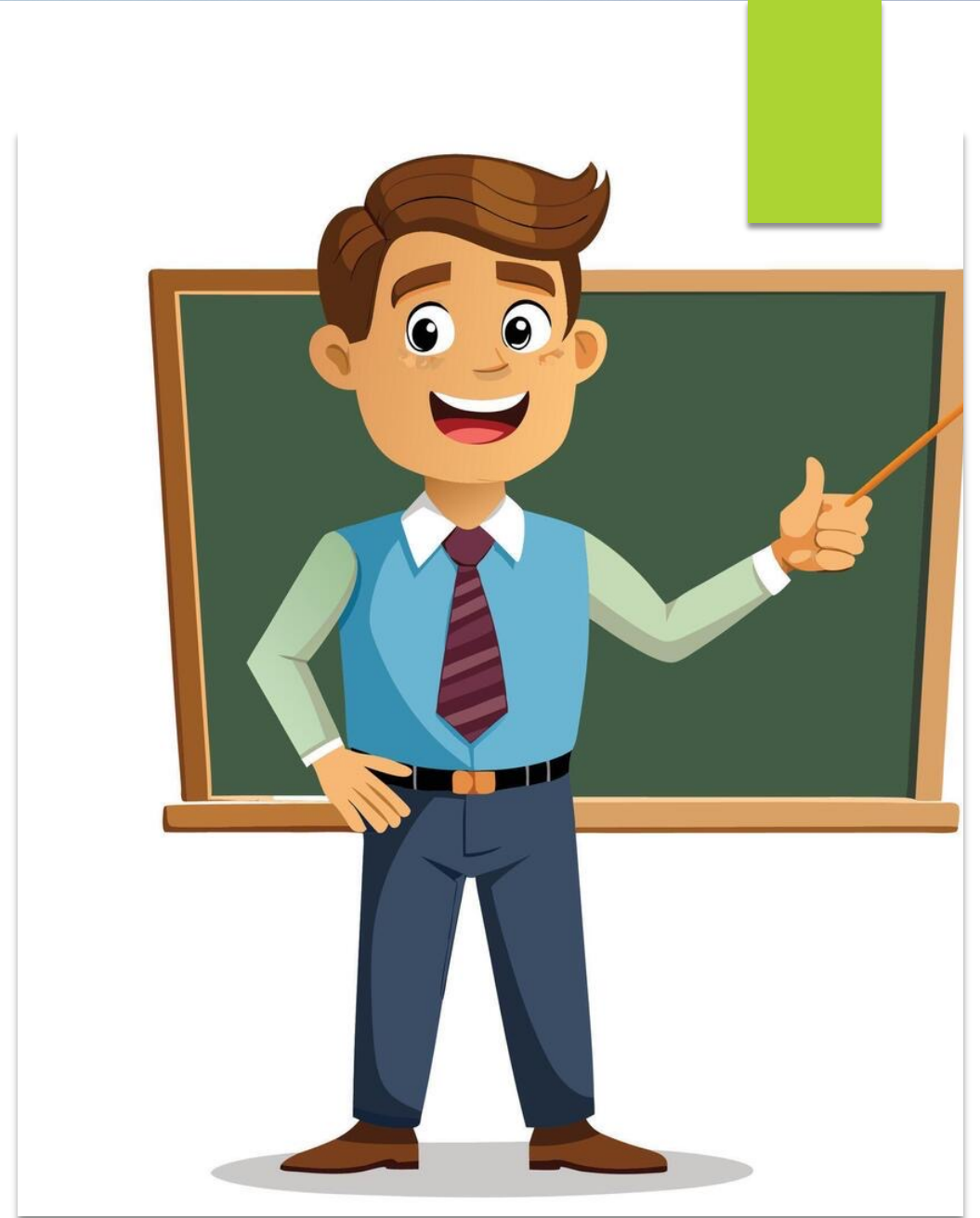
Tell AI who to be:

"You are a history teacher..."

"You are a fun inventor..."

"You are a kind coach..."

Good roles, better tone ideas!



Technique 2 – Multi Step Prompts

Break big tasks into steps:

- Plan
- Create
- Review

Example:

"First, list 3 ideas for a school recycling campaign. Then choose the best one. Finally, write a poster slogan."



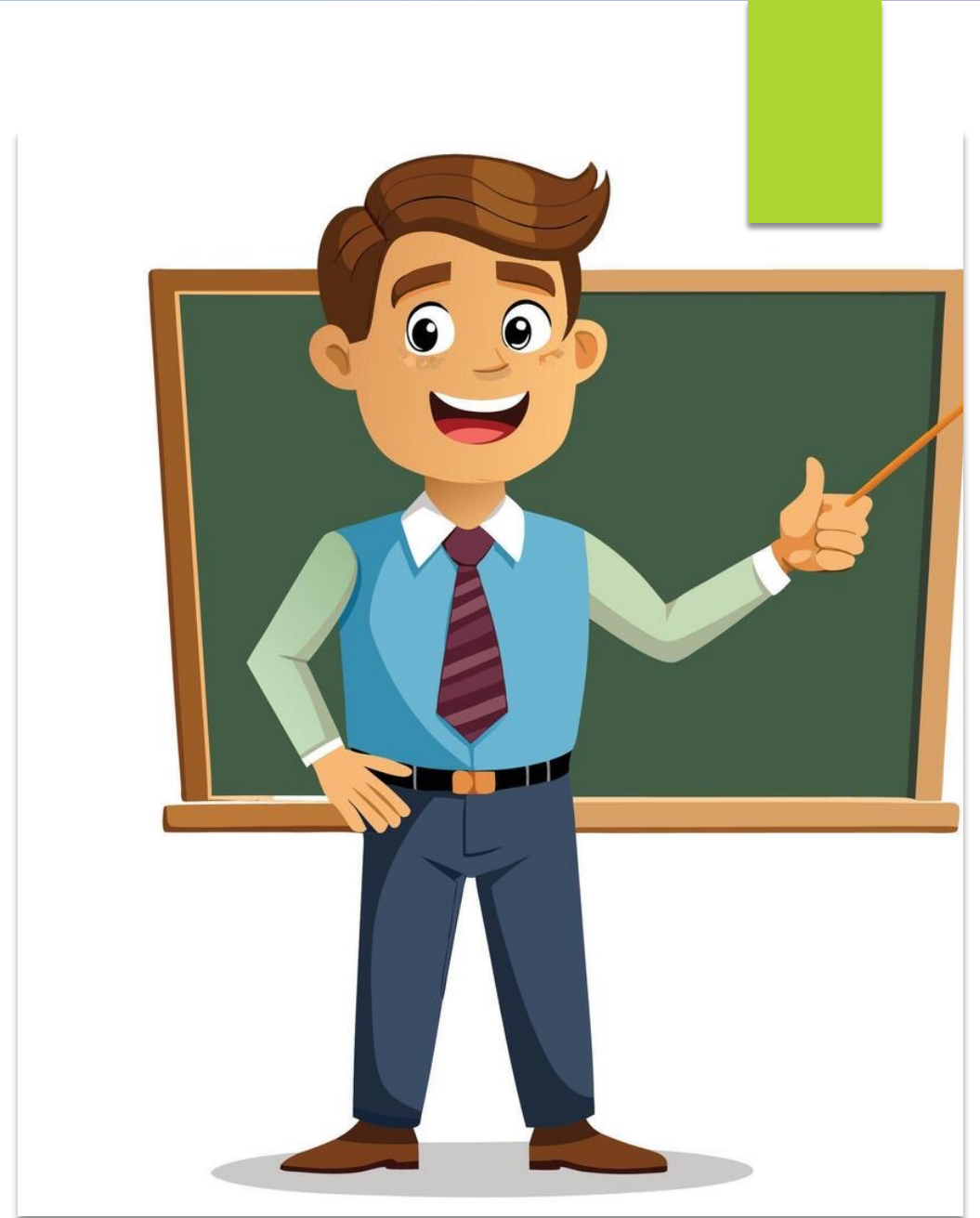
Technique 3 – Iteration (Improve it!)

You don't have to stop at the first answer!
Improve it by giving feedback to the AI

Examples:

"Make it shorter and funnier"

"Add examples for Year 7 students"



Activity – Prompt makeover challenge

AI won't always get it right the first time, and that's okay!

In this activity, you will practice iteration, which means improving your prompts through feedback and testing

Start with this simple prompt: "Write about friendship"

Improve it 3 times (make it clearer each round)

Share your best version!





End of
Lesson 3!

Lesson 4 - Responsible and Ethical AI Use

INPUT PROMPT

You are a helpful AI assistant.
Explain quantum computing
in simple terms.
Include an example.

84 / 1000

MODE

Conversational

tone

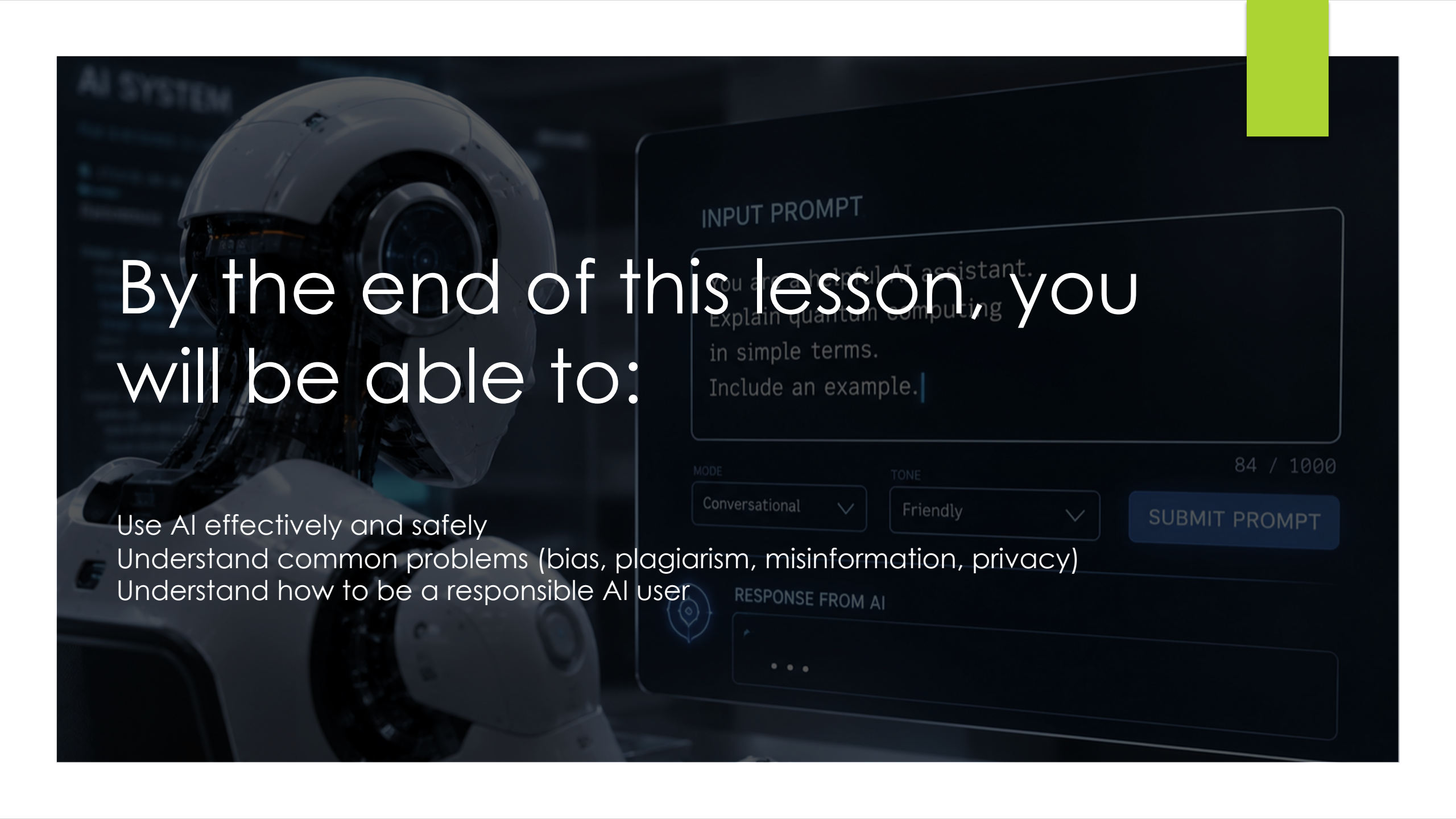
Friendly

SUBMIT PROMPT



RESPONSE FROM AI

...

The background of the slide features a futuristic robot head on the left, partially obscured by the text. On the right, a computer screen displays an AI interface. The screen has a dark theme with light blue and white text. At the top, it says 'INPUT PROMPT'. Below that, there is a text box containing the prompt: 'You are a helpful AI assistant. Explain quantum computing in simple terms. Include an example.' Below the text box, there are two dropdown menus labeled 'MODE' and 'TONE'. The 'MODE' dropdown is set to 'Conversational' and the 'TONE' dropdown is set to 'Friendly'. To the right of these dropdowns, it says '84 / 1000'. Below the dropdowns is a blue button labeled 'SUBMIT PROMPT'. At the bottom of the screen, there is a section labeled 'RESPONSE FROM AI' with a loading indicator (three dots) and a small icon of a robot head.

By the end of this lesson, you will be able to:

- Use AI effectively and safely

- Understand common problems (bias, plagiarism, misinformation, privacy)

- Understand how to be a responsible AI user

The 4 biggest things to watch out for

Bias – AI can copy unfair ideas from its training data

Plagiarism – Don't copy AI work as your own

Misinformation – AI can make mistakes or invent facts

Privacy – Don't share personal information



Final Activity – Would you risk it?

This is a mini ethical decision game. There is no right or wrong – just your judgment.

Instructions:

Read each scenario on the next slide

Choose if it is okay to use AI or not okay to use AI

Write a one sentence explanation of why

Discuss your reasoning with a partner or a small group



Final Activity – Would you risk it?

Scenarios

Your group decides to use AI to generate half of a report to save time, but only one member plans to declare it in the submission notes.

After receiving teacher feedback, you paste your whole essay into ai and ask it to "rewrite based on this feedback" instead of revising sections yourself.

You use AI to help check grammar and structure before submission.

Before an exam, you use AI to practice by asking it to quiz you with sample questions based on your notes. You double-check the answers yourself afterward.

You use AI to summarise lecture notes you already wrote yourself

